



4.2 Summarising data: measures and displays

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Central tendency is measured by the mean, median, or mode; spread by the range, interquartile range ($IQR=Q3-Q1$), variance, or standard deviation. Box-and-whisker plots display the five-number summary (minimum, $Q1$, median, $Q3$, maximum); histograms and cumulative frequency graphs summarise grouped data and allow the median/quartiles to be estimated. The mean is sensitive to outliers, while the median is more resistant, making the median often preferable for skewed data.

Worked example: follow the reasoning

The times (minutes) for 7 runners are 22, 25, 23, 30, 24, 26, and 80 (an injured runner). Find the mean and median, and comment on which is more representative.

Solution: Sorted: 22, 23, 24, 25, 26, 30, 80. Median = 25 (4th value). Mean = $(22+23+24+25+26+30+80)/7 = 230/7 \approx 32.86$. The outlier (80) inflates the mean well above most values; the median (25) better represents a typical runner's time.

Quick check · multiple choice

A data set has $Q1=12$ and $Q3=20$. What is the interquartile range?

- A 8
- B 16
- C 32
- D 4

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The grouped frequency table shows delivery times (minutes) for 60 orders: [6]
- [10,20): 8, [20,30): 18, [30,40): 22, [40,50): 12. (a) Estimate the mean delivery time using midpoints. (b) State the modal class. (c) State which class interval contains the median, explaining your reasoning using cumulative frequencies.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Midpoints 15,25,35,45: $\text{mean} = (8 \times 15 + 18 \times 25 + 22 \times 35 + 12 \times 45) / 60 = 1880 / 60 \approx 31.33$ minutes.
(b) Modal class: [30,40) (highest frequency, 22). (c) Cumulative frequencies are 8, 26, 48, 60; the median position is the 30th value, and since the cumulative frequency reaches 26 by the end of [20,30) and 48 by the end of [30,40), the 30th value lies within [30,40).

Quick check answer

A. $\text{IQR} = Q_3 - Q_1 = 20 - 12 = 8$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.